

Apex Bank Microservices “Clone & Setup Guide

Complete instructions for running the distributed banking platform on a new laptop

GITHUB REPOSITORY

Saisanjay1410/Apex-Bank-Microservices

TARGET PLATFORM

Windows 10/11 / macOS / Linux

DEFAULT GATEWAY PORT

Port 8090 (Frontend: 5173)

1. Prerequisites to Install on the New Laptop

Ensure the following software packages are installed on the target computer prior to running:

Tool	Required Version	Verification Command	Installation Source
Git	v2.30 or newer	<code>git --version</code>	git-scm.com/downloads
Java JDK	JDK 17 or JDK 21	<code>java -version</code>	adoptium.net (Eclipse Temurin)
Node.js & npm	Node v18+ / v20+	<code>node -v & npm -v</code>	nodejs.org
MySQL Server	MySQL 8.0+	<code>mysql --version</code>	dev.mysql.com/downloads/installer
Apache Maven	v3.8+ (Optional)	<code>mvn -v</code>	Included Maven Wrapper (./mvnw) is provided

2. Clone the Repository

Open a terminal (PowerShell, Command Prompt, or Terminal) in your desired working folder and clone:

```
# Clone the repository from GitHub
git clone https://github.com/Saisanjay1410/Apex-Bank-Microservices.git

# Enter the project directory
cd Apex-Bank-Microservices
```

3. MySQL Database Configuration

1. Start the MySQL service and open the MySQL Command Line Client or Workbench.
2. Execute the following SQL query to create the database:

```
CREATE DATABASE IF NOT EXISTS banking_payroll;
```

No Manual Schema Required: Hibernate DDL auto-creates all tables (users, roles, bank_account, transaction, employee) and seeds default users on first start.

Database Password Notice: The microservices default to password Sanjay@2004. If your local MySQL root password differs, update `spring.datasource.password` in:

- `Banking-Service/src/main/resources/application.properties`

- `Payroll-Service/src/main/resources/application.properties`

Alternatively, pass your password at launch time as shown in Step 4.

4. Step-by-Step Execution Sequence (6 Terminals)

Open separate terminal windows within the Apex-Bank-Microservices root directory and run the services in the following order:

Terminal 1: Eureka Discovery Registry

PORT 8761

The central registry must be started first so that downstream services can register themselves.

```
cd Eureka-Server
mvn spring-boot:run # Or .\mvnw.cmd spring-boot:run
```

Verification: Wait ~15s, then visit <http://localhost:8761> in your browser.

Terminal 2: AI Assistant Microservice

PORT 8083

Provides conversational intelligence and answers user queries regarding accounts and payroll.

```
cd AI-Chatbot_Geethika
mvn spring-boot:run
```

Terminal 3: Core Banking Microservice

PORT 8081

Manages user authentication, accounts, transaction ledgers, and atomic fund transfers.

```
cd Banking-Service
mvn spring-boot:run
# If overriding password: mvn spring-boot:run -Dspring-boot.run.arguments="--
spring.datasource.password=YOUR_PASS"
```

Terminal 4: Payroll & Employee Microservice

PORT 8082

Handles employee registry, department management, and automated salary disbursements.

```
cd Payroll-Service
mvn spring-boot:run
```

Terminal 5: Spring Cloud API Gateway

PORT 8090

Routes all inbound client requests, manages CORS, and forwards to registered Eureka instances.

```
cd API-Gateway
mvn spring-boot:run
```

Configured to port 8090 to avoid default conflicts with Oracle XETNS or IIS on 8080.

Terminal 6: React Frontend Portal

PORT 5173

Modern React 18 single-page application built with TypeScript, Tailwind CSS, and Vite.

```
cd frontend
npm install
npm run dev
```

Access Portal: Open your browser to <http://localhost:5173>

5. Seed User Login Credentials

Log in at <http://localhost:5173/login> with any of the pre-configured accounts:

Role	Username	Password	Accessible Features
Administrator	admin	admin123	Analytics Dashboard, All Bank Accounts, Complete Audit Ledger, User Roles
HR Manager	hr_user	hr123	Payroll Processing, Employee Directory, Salary Runs, AI Assistant
Employee	emp_user	emp123	Personal Banking Portal, Funds Transfer, Salary Slips, AI Assistant

6. Common Troubleshooting FAQs

- Port 8080 Already in Use:** Apex Bank API Gateway runs on port 8090, intentionally bypassing any existing Apache Tomcat or Oracle services.
- MySQL Connection Refused:** Verify the MySQL 8.0 service is running via Windows Services (Start-Service MySQL80) or `sudo systemctl start mysql` on Linux.
- Instance Not Visible in Eureka:** Allow 10-20 seconds after starting a service for the heartbeat signal to register with Eureka at `http://localhost:8761`.